

Automated Scoring of Speech

11.1 INTRODUCTION

We have so far considered automated scoring of *written* responses. A neighboring field of automated scoring of *spoken* responses shares certain characteristics with our primary field of interest; we therefore provide a brief overview. We refer a reader interested in learning more to the recently published edited volume on automated assessment of speech (Zechner & Evanini, 2020); the discussion in this chapter is informed by that work.

Spoken responses in assessment tasks range from *scripted* speech (that is, reading out loud or repetition of the prompt) to *spontaneous* speech, namely, responses to open-ended questions such as “Describe your recent vacation”. Intermediate locations on the scripted-vs.-spontaneous spectrum are also possible, for example, a task with moderately predictable speech where the test-taker produces a sentence using certain key words, or a picture narration task where the target vocabulary is somewhat restricted to the detail of the picture (Loukina, Zechner, et al., 2017).

Early work on speech scoring concentrated on scripted speech and, therefore, on the accuracy of the content (the extent to which what the test-taker said matches the prompt) and on its DELIVERY (fluency and pronunciation; e.g., overall pronunciation accuracy, number of disfluencies, stress patterns, intonation). Both of these aspects of speech production are assessed in moderately open-ended tasks as well; measurement of accuracy becomes somewhat more involved, since there is no one correct answer and many close variants are acceptable. Moderately open-ended tasks could also support measurement of what is called *language use*—typically covering vocabulary choice and grammar.

For spontaneous speech, and especially tasks that elicit extended spoken responses, delivery and language use elements in the scoring rubric are often joined by discourse and topic development considerations. One can clearly perceive parallels to the essay scoring world, with grammar, pronunciation, and prosody conventions (of spoken, rather than written, English), vocabulary, and discourse-level aspects of a spoken composition being assessed.

The scripted-vs.-spontaneous dimension is relevant for the design of the speech scoring system for at least two reasons. The first is similar to the distinction between scoring for content and scoring for proficiency that we discussed in the context of written responses (see Chapter 10). The more open-ended the question, the harder it is to operationalize the notion of a *correct* response in terms of content accuracy; on the other hand, open-ended tasks allow test-takers to demonstrate proficiency in terms of use of nuanced vocabulary, topic development, coherence,

etc. The second reason is more specific to the speech scoring pipeline—the first step in automated scoring of speech is automated recognition of the words uttered by the test-taker, and the larger the possible vocabulary, the harder the task.

In this chapter, we start with reviewing the typical components of an automated speech recognition system (ASR)—a foundational building block of speech scoring systems. We then proceed to review the main elements of a system for scoring spontaneous speech, using research on SpeechRater®—a state of the art system for scoring spontaneous speech. We review features capturing aspects of delivery, vocabulary, and discourse, and draw parallels to the text-based work we discussed in the rest of the book.

11.2 AUTOMATED SPEECH RECOGNITION FOR SPEECH SCORING

Automated speech recognition (ASR) is a process that converts the speech signal in a digital audio recording into text. The speech signal—voiced sounds, unvoiced sounds, and silences—is measured at regular short intervals across various frequencies. These measurements serve as the input into an ACOUSTIC MODEL, whose job it is to translate them into phonemes or sequences of phonemes. Acoustic models are typically implemented using a hybrid Deep Neural Network and Hidden Markov Model approach (DNN-HMM).

The next component of an ASR system is a *pronunciation dictionary* that maps the orthographic form of a word to its sequence of phonemes. The words listed in the pronunciation dictionary are the words that the ASR system can recognize.

The components mentioned so far can generate a candidate sequence of words based on the acoustic signal; a *language model* helps pick the likelier sequences. Language models are usually implemented as n-gram models, with a trigram model being the most popular.

The process of speech recognition is a search for the most probable sequence of words uttered in the given recording, given the acoustic measurements, the acoustic model, the pronunciation dictionary, and the language model; we note that some of the current end-to-end systems do not explicitly contain all of these components.

The standard measure used to evaluate an ASR system is WORD ERROR RATE (WER)—the discrepancy between the ASR-hypothesized sequence of words and that provided by a human transcriber. The ASR hypothesis and the transcription are first aligned using dynamic programming, minimizing the *edit distance* between the hypothesis and the transcription—the number of edits (insertions, deletions, or substitutions) needed to change one string into the other. Using the optimal alignment, three types of ASR errors are recorded: (a) insertion (ASR produced a word where no word was transcribed), (b) deletion (ASR produced no word where a word was transcribed), and (c) substitution (ASR and transcription have different words). The WER is the sum of the counts of the three types of errors divided by the number of words in the transcription; the lower the value, the better the system's performance.